Chapter 1 The Space L^p and Spaces of Measures

1.1 Space L^p

Definition 1.1.0.1 Let $(X, \mathcal{A}, \mu)$ be a measure space, with nonegative measure μ, and $p \in [1, \infty)$. Then denote

$$\mathcal{L}^p(\mu) := \{f : X \to [-\infty, \infty] | |f|^p \text{ is } \mu\text{-integrable}\}$$

Define μ-equivalent: The sets $\mathcal{L}^p(\mu)$ are equipped with the equivalent relation, μ-almost everywhere equivalent

$$f \sim g \Leftrightarrow \mu(\{x \in X | f(x) \neq g(x)\}) = 0$$

Then, denote $L^p(\mu)$ as the factor-space of $\mathcal{L}^p(\mu)$ with respect to the equivalence relation.
The norm structure is specified by

$$\| - \|_{L^p(X,\mu)} : f \mapsto (\int_X |f|^p d\mu)^{\frac{1}{p}}$$

Furthermore, with the limit

$$\lim_{p\to\infty} (\int_X |f|^p d\mu)^{\frac{1}{p}}$$

Norm $\| \cdot \|_p$

◇ **Lemma 1.1.0.2** For all $\lambda \in \mathbb{R}$, $f, g \in \mathcal{L}^p(\mu)$, we have

$$\|\lambda f\|_p = |\lambda| \|f\|_p \qquad \|f + g\|_p \leq \|f\|_p + \|g\|_g$$

∘ **Proof** Easy to check, by property of integration. □

★ **Corollary 1.1.0.3** The function $\| \cdot \|_p$ is a norm on the space $L^p(\mu)$.

∘ **Proof** Omitted. □

▶ ***Theorem*** The space $L^p(\mu)$ is compelete, $p \in [1, +\infty]$.

∘ **Proof** Consider the Cauchy sequence □

1.1.1 Convolutions

Definition 1.1.1.1 Suppsoe $f, g \in L^1(G)$. Define the **convolution**

$$f * g(x) := \int_G f(y)g(y^{-1}x)d\lambda(y)$$

Especially, when $G = (\mathbb{R}^n, +)$, then

$$f * g(x) = \int_{\mathbb{R}^n} f(y)g(x-y)dy$$

Especially, change of variable $z = x^{-1}y$,

$$= \int_G f(xz)g(z^{-1})d\lambda(z)$$

♠ **Proposition 1.1.1.2** For all $f, g, h \in L^1(G)$

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∘ **Proof** ... □

Convolution Inequalities

▶ ***Theorem*** (Minkowski with Convolution)

$$\|g * f\|_{L^p(G)} \leq \|g\|_{L^1(G)}\|f\|_{L^p(G)}$$

∘ **Proof** ... □

▶ ***Theorem*** (Young)

∘ **Proof** ... □

▶ ***Theorem*** (Young)

∘ **Proof** ... □

Approximate Identities

Definition 1.1.1.3 ...

Definition 1.1.1.4 Let T be an operator defined on a linear space of complex-valued measure functions on a measure space

Then, T is called **linear** if for all $f, g \in T$, and $\lambda \in \mathbb{C}$, we have

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1.1.2 Properties

Definition 1.1.2.1 For $0 < p < \infty$, the space $L^p_{loc}(\mathbb{R}^n, |\cdot|)$ or simply $L^p_{loc}(\mathbb{R}^n)$ is the set of all Lesbegue0mesurable functions f on $\mathbb{R}^n$, that satisfies

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is the **locally p-integrable function** on $\mathbb{R}^n$.

▶ ***Theorem*** (Hölder) Let $0 < p, p_1, \dots, p_k \leq \infty$, where $k \geq 2$, and let f_j be

▶ ***Theorem*** (Jensen)

▶ ***Theorem***

- (Discrete Minkowski)
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  - –
- (Integral Inequality)

∘ **Proof** ... □

▶ ***Theorem*** (Young)

▶ ***Theorem*** (Hardy's Inequality)

1.1.3 Approximation of L^p

◇ **Lemma 1.1.3.1** The set of all simple functions is everywhere dense in every space $L^p(\mu), 1 \leq p < \infty$.

∘ **Proof** ... □

★ **Corollary 1.1.3.2** Let a nonnegative Borel measure μ on $\mathbb{R}^n$ be bounded on bounded sets.
Then, the class $C_0^\infty(\mathbb{R}^n)$ of smooth functions with bounded support is everywhere dense in $L^p(\mu), 1 \leq p < \infty$.
In particular, the spaces $L^p(\mu), 1 \leq p < \infty$, are separable.

∘ **Proof** ... □

Definition 1.1.3.3 Let ρ be a function integrable over $\mathbb{R}^n$ such that

$$\int_{\mathbb{R}^n} \rho(x)dx = 1$$

Now, set $\rho_\epsilon(x) = \frac{\rho(\frac{x}{\epsilon})}{\epsilon^n}, \epsilon > 0$.

◇ **Lemma 1.1.3.4** Let $f \in \mathcal{L}^p(\mathbb{R}^n), 1 \leq p < \infty$, then the mapping

$$T_f : \mathbb{R}^n \to L^p(\mathbb{R}^n) \qquad T_f(v)(x) = f(x+v)$$

is continuous and bounded.

∘ **Proof** ... □

▶ ***Theorem*** Let $f \in \mathcal{L}^p(\mathbb{R}^n), 1 \leq p < \infty$, then one has

$$\lim_{\epsilon \to 0} \|f * \rho - f\|_p = 0$$

∘ **Proof** ... □

★ **Corollary 1.1.3.5** ...

∘ **Proof** ... □

1.1.4 The Weak Topology in L^p

1.2 Uniform Integrability

1.3 Convergence of Measures

1.4 The Hilbert Space L^2

Definition 1.4.0.1 Let μ be a measure with values in $[0, +\infty]$.
The space $L^2(\mu)$ is distinguished among other $L^p(\mu)$ by the property that it is

- Euclidean: With the inner product

$$(f, g) = \int_X f g d\mu$$

- or, Hermitian: With the inner product

$$(f, g) = \int_X f \bar{g} d\mu$$

Notice: From $|fg| \leq f^2 + g^2$,

$$f, g \in \mathcal{L}^2(\mu) \Rightarrow fg \in \mathcal{L}^1(\mu)$$

♠ **Proposition 1.4.0.2** Let H_0 be a closed linear subspace in a Hilbert space.
Then, $H_0^{\perp} = \{x \in H | x \perp h, \forall h \in H_0\}$ is a closed linear subspace in H, and

$$H = H_0 \oplus H_0^{\perp}$$

Hence, for every $h \in H$, there exists a unique vector h_0, with $h - h_0 \in H_0^{\perp}$. In addition

$$\|h - h_0\| = \inf\{\|h - x\| | x \in H_0\}$$

∘ **Proof** ... □

★ **Corollary 1.4.0.3** Let f be a continuous linear function on a Hilbert space H.
Then, there exists a unique vector v such that

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∘ **Proof** ... □

★ **Corollary 1.4.0.4**

- There exists a family of mutually orthogonal unit vector $e_\alpha \in L^2(\mu)$ such that: Every element f in $L^2(\mu)$ is the sum of the following series covergent in $L^2(\mu)$:

  $$f = \sum_\alpha c_\alpha e_\alpha$$

  where at most countably many coefficients c_α may be nonzero.

  In addition, one has

  $$c_\alpha = (f, e_\alpha) \qquad \|f\|^2 = \sum_\alpha |c_\alpha|^2$$

  The family is calleld an **orthogonal basis** of space $L^2(\mu)$.

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∘ **Proof** ... □

▶ ***Theorem*** For any orthonormal system $\{\varphi_n\}$ in $L^2(\mu)$, and any sequence $\{c_n\} \in l^2$, the series

$$\sum_{n=1}^{\infty} c_n \varphi_n \in L^2(\mu)$$

∘ **Proof** ... □

▶ ***Theorem*** Let $\{\varphi_n\}$ be an orthonormal sequence $L^2(\mu)$.
Then, for all $f \in L^2(\mu)$, the following Bessel inequality holds

$$\sum_{n=1}^{\infty} |(f, \varphi_n)|^2 \leq \|f\|^2_{L^2(\mu)}$$

In particular, this equality is true if $\{\varphi_n\}$ is an orthonormal basis.

∘ **Proof** ... □

1.5 Duality of the Space L^p

Chapter 2 Weak L^p Spaces

2.1 Definitions and Properties

Definition 2.1.0.1 Take the following of distribution function

$$d_f(\alpha) := \mu(\{x \in X || f(x) > \alpha\}) \qquad \alpha > 0$$

Let $f : X \to \mathbb{C}$ be measurable, with $p \in [1, \infty)$, then define

$$\|f\|_{L^{p,\infty}} := \inf\{C > 0 | d_f(\alpha) \leq (\frac{C}{\lambda})^p, \forall \alpha > 0\} = \sup_{\lambda > 0} \lambda d_f(\lambda)^{\frac{1}{p}}$$

2.1.1 L^p and $L^{p,\infty}$

♠ **Proposition 2.1.1.1** There is embedding

$$\mathcal{L}^p(X, \mu) \hookrightarrow \mathcal{L}^{p,\infty}(X, \mu)$$

∘ **Proof** ... □

2.1.2 Covergence in Measure

Chapter 3 Lorentz Spaces

3.1 Definitions

Definition 3.1.0.1 ...

3.1.1 Duals of Lorentz Spaces

Chapter 4 Interpolation Theory of Lesbegue Space

In general, we will develope the interpolation theory in Banach space. Here we just mainly concern

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♠ **Proposition 4.0.0.1** (Interpolation in Lesbegue Space)Let $0 < p < q \leq \infty$, and let $f \in L^{p,\infty}(X,\mu) \cap L^{q,\infty}(X,\mu)$, where X is a σ-finite measure space.

Then, $f \in L^r(X,\mu)$, for all $p < r < q$ and

$$\|f\|_{L^r} \leq (\frac{r}{r-p} + \frac{r}{q-r})^{\frac{1}{r}} \|f\|_{L^{p,\infty}}^{\frac{\frac{1}{r}-\frac{1}{q}}{\frac{1}{p}-\frac{1}{q}}} \|f\|_{L^{q,\infty}}^{\frac{\frac{1}{p}-\frac{1}{r}}{\frac{1}{p}-\frac{1}{q}}}$$

∘ **Proof** ... □

4.1 Real Interpolation

4.2 Complex Interpolation

4.3 Interpolation of Operators

4.3.1 Interpolation of Analytic Family of Operators